

Benjamin Rommelaere

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Citizenship:

Canadian

Research Interests:

Environmental Economics, Industrial Organization,
Applied Microeconomics

EDUCATION

Ph.D. in Economics, University of Toronto 2019-2026 (Expected)

Committee: Eduardo Souza-Rodrigues (supervisor), Victor Aguirregabiria,
Stephan Heblich, Adonis Yatchew

Parental leave: May–Aug 2022; Jan–Apr 2025

M.A. in Economics, Queen's University 2013

B.A. in Economics (Honours), Toronto Metropolitan University 2012

WORKING PAPERS

The Climate Impact of Wood Bioenergy Subsidies: A Structural Model of Forest Land Use (Job Market Paper)

Market Concentration and Deforestation: Evidence from the Brazilian Soy Industry

WORK-IN-PROGRESS

Incentives and Corruption in Panama: Evidence from Gasoline Markets, with Steven Lehrer and Decio Coviello

Environmental Regulation of Wood Processing Mills: The Boiler MACT Rule

Optimal Spatial Targeting of Wood Bioenergy Subsidies

CONFERENCE PROCEEDINGS

Making Transit Reliability Benefits Accessible to Engineers, with John Parker
Proceedings of the International Conference on Transportation and Development, 2016.

AWARDS AND GRANTS

SSHRC Doctoral Fellowship (\$40,000 x 2) 2023-2024

CPE Climate Solutions Scholarship (\$15,000) 2023

University of Toronto Doctoral Fellowship (\$20,000 x 5) 2019-2023

Queen's University Graduate Scholarship (\$17,500) 2013

Toronto Metropolitan University's Best Thesis Award (\$5,000) 2013

PROFESSIONAL EXPERIENCE

Teaching Assistant, *University of Toronto*

2019–present

- ECO439: Empirical Methods in Microeconomics
- ECO414 / ECO1960: Energy and Regulation
- ECO403: Topics in Development Economics and Policy
- ECO314: Energy and the Environment
- ECO313: Environmental Economics and Policies
- ECO227: Foundations of Econometrics
- ECO220: Introduction to Data Analysis and Applied Econometrics
- ECO202: Macroeconomic Theory and Policy
- ECO101: Principles of Microeconomics
- ENV462: Energy and Environment – Economics, Politics, and Sustainability
- ENV347: The Power of Economic Ideas

Consulting Associate, *Charles River Associates*

2017–2019

- Economics research in support of antitrust litigation and regulatory board hearings

Economist, *Autocase Economic Advisory*

2014–2017

- Economics research in support of infrastructure investment decisions for government agencies

Data Analyst, *Jany Analytics*

2014

Research Analyst, *AOL Canada*

2012

Research Assistant, Prof. Paul Missios, *Toronto Metropolitan University*

2011

PROFESSIONAL SERVICE

Assistant to the Editor-in-Chief, *The Energy Journal* (referee selection)

2023–2024

MAJOR PRESENTATIONS

2025: University of Toronto, Climate Economics Seminar; Rotman School of Management, Economics and Analysis Program Seminar

LANGUAGE & TECHNICAL SKILLS

English (native)

Programming: Python, R, Stata, MATLAB, ArcGIS

REFERENCES

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Abstracts

The Climate Impact of Wood Bioenergy Subsidies: A Structural Model of Forest Land Use

(Job Market Paper)

Wood bioenergy has become an important renewable energy source in the EU and UK, driven largely by generous subsidies. Yet burning wood emits more CO₂ than coal at the smokestack, so any climate benefit depends on whether the resulting change in forest carbon stocks offsets these emissions. I study the effects of these subsidies on forests in the U.S. South—the world’s main export region for wood bioenergy—and find the opposite: the forest carbon sink weakens, reflecting reduced sequestration, making the policy counterproductive both at the point of emissions and in the forest. To identify these effects, I develop a dynamic structural model combining land-use and harvesting decisions under local oligopsony power to capture the trade-off landowners face between harvesting, costly replanting, and alternative land uses. The model is estimated on a panel of 5.1 million land plots built from remote-sensing data on land use, harvesting, and biomass. Results show that policy-induced harvesting is not offset by increases in planting, leading to net deforestation and declining forest carbon stocks. The impact is substantial: in 2024, the estimated reduction in annual carbon sequestration equals 1.4 percent of U.K. emissions, and by 2050, the social cost of lost forest carbon is projected to reach \$53 billion. The analysis suggests that location-based sourcing restrictions could reverse this negative impact and deliver social benefits instead.

Market Concentration & Deforestation: Evidence from the Brazilian Soy Industry

The Brazilian soy industry is a leading cause of deforestation in several major biomes, including the Amazon, yet little is known about how market structure affects land-use outcomes. This paper exploits the 2014 acquisition and merger of two major soy exporters by China’s state-owned firm COFCO to estimate the causal effects of buyer concentration on farmgate soy prices, production, and deforestation. Using a municipality-level panel from 2006–2018 that links soy market supply-chain data, administrative data on prices and production, and MapBiomass deforestation records, I implement a staggered event-study design to estimate local impacts of the merger. Results show a short-run increase in farmgate prices where buyer concentration rose — an unexpected effect consistent with strategic mutual forbearance among oligopsonists. In contrast, when COFCO enters new markets, I find that increased competition leads to sustained price increases and lower deforestation. One likely mechanism is COFCO’s deforestation-free sourcing commitments.

Incentives and Corruption in Panama: Evidence from Gasoline Markets

with Decio Coviello & Steven Lehrer

This paper studies the misuse of public funds using transaction-level data from Panama’s national fuel card program, covering over two million purchases by government employees. We document both passive waste, arising from weak cost-minimization incentives, and active misuse, where employees personally benefit. We use the fuel programs rules to develop waste and misuse measures and exploit two distinct sources of variation in the incentives to misuse funds. A local price shock from a merger involving the contracted fuel supplier reduces misuse across most agencies, whereas, national fuel price shocks driven by exogenous oil market events increase it. We develop an economic model of expected profit and detection risk to explain these behavioral responses and their heterogeneity across agencies. Our estimates imply an elasticity of supply of stolen fuels between 2.7 and 4.6, meaning that a 10% rise in fuel prices increases stolen fuel by 27–46%.